



National  
Qualifications  
RESOURCE

---

**2021 Physics  
Section 1  
  
Higher  
  
Finalised Marking Instructions**

© Scottish Qualifications Authority 2021

These marking instructions have been prepared by examination teams for use by SQA appointed markers when marking external course assessments.

The information in this document may be reproduced in support of SQA qualifications only on a non-commercial basis. If it is reproduced, SQA must be clearly acknowledged as the source. If it is to be reproduced for any other purpose, written permission must be obtained from [permission@sqa.org.uk](mailto:permission@sqa.org.uk).



## Marking instructions for each question

| Question | Answer | Mark |
|----------|--------|------|
| 1.       | B      | 1    |
| 2.       | E      | 1    |
| 3.       | A      | 1    |
| 4.       | A      | 1    |
| 5.       | A      | 1    |
| 6.       | C      | 1    |
| 7.       | C      | 1    |
| 8.       | D      | 1    |
| 9.       | C      | 1    |
| 10.      | B      | 1    |
| 11.      | D      | 1    |
| 12.      | B      | 1    |
| 13.      | B      | 1    |
| 14.      | D      | 1    |
| 15.      | E      | 1    |
| 16.      | B      | 1    |
| 17.      | E      | 1    |
| 18.      | D      | 1    |
| 19.      | A      | 1    |
| 20.      | D      | 1    |
| 21.      | D      | 1    |
| 22.      | C      | 1    |
| 23.      | A      | 1    |
| 24.      | C      | 1    |
| 25.      | D      | 1    |

[END OF MARKING INSTRUCTIONS]

The following table provides information on each question including: Course content being assessed, Skills assessed (see Physics Understanding Standards materials for a definition of each code); Maximum Mark; A-type marks.

| Question | Part | Course content                                                     | Skills assessed | Maximum mark | A-type marks |
|----------|------|--------------------------------------------------------------------|-----------------|--------------|--------------|
| 1        |      | Our dynamic Universe - motion-equations and graphs                 | K3              | 1            |              |
| 2        |      | Our dynamic Universe - motion-equations and graphs                 | K1              | 1            |              |
| 3        |      | Our dynamic Universe - motion-equations and graphs                 | K3              | 1            |              |
| 4        |      | Our dynamic Universe - forces, energy, and power                   | K3              | 1            | 1            |
| 5        |      | Our dynamic Universe - forces, energy, and power                   | S6              | 1            |              |
| 6        |      | Our dynamic Universe - forces, energy, and power                   | K3              | 1            |              |
| 7        |      | Our dynamic Universe - collisions, explosions, and impulse         | K3              | 1            |              |
| 8        |      | Our dynamic Universe - skills                                      | S4              | 1            |              |
| 9        |      | Our dynamic Universe - special relativity                          | K1              | 1            |              |
| 10       |      | Our dynamic Universe - special relativity                          | K3              | 1            |              |
| 11       |      | Our dynamic Universe - expanding Universe                          | K3              | 1            | 1            |
| 12       |      | Our dynamic Universe - expanding Universe                          | K1              | 1            |              |
| 13       |      | Particles and waves - forces on charged particles                  | S6              | 1            | 1            |
| 14       |      | Particles and waves - Standard Model                               | K1              | 1            |              |
| 15       |      | Particles and waves - Standard Model                               | S4              | 1            | 1            |
| 16       |      | Particles and waves - nuclear reactions                            | K3              | 1            |              |
| 17       |      | Particles and waves - wave-particle duality                        | S5              | 1            |              |
| 18       |      | Particles and waves - interference                                 | K3              | 1            | 1            |
| 19       |      | Particles and waves - refraction of light                          | K3              | 1            |              |
| 20       |      | Particles and waves - refraction of light                          | S6              | 1            | 1            |
| 21       |      | Electricity - current, potential difference, power, and resistance | K3              | 1            |              |
| 22       |      | Electricity - current, potential difference, power, and resistance | K3              | 1            |              |
| 23       |      | Electricity - capacitors                                           | K1              | 1            |              |
| 24       |      | Electricity - capacitors                                           | K3              | 1            |              |
| 25       |      | Electricity - capacitors                                           | S6              | 1            |              |



National  
Qualifications  
RESOURCE

---

**2021 Physics  
Section 2  
  
Higher  
  
Finalised Marking Instructions**

© Scottish Qualifications Authority 2021

These marking instructions have been prepared by examination teams for use by SQA appointed markers when marking external course assessments.

The information in this document may be reproduced in support of SQA qualifications only on a non-commercial basis. If it is reproduced, SQA must be clearly acknowledged as the source. If it is to be reproduced for any other purpose, written permission must be obtained from [permission@sqa.org.uk](mailto:permission@sqa.org.uk).



## General marking principles for Physics Higher

Marks for each candidate response must always be assigned in line with these marking principles, the Physics: general marking principles (GMPs)

([http://www.sqa.org.uk/files\\_ccc/Physicsgeneralmarkingprinciples.pdf](http://www.sqa.org.uk/files_ccc/Physicsgeneralmarkingprinciples.pdf)) and the detailed marking instructions for this assessment.

- (a) Marking should always be positive. This means that, for each candidate response, marks are accumulated for the demonstration of relevant skills, knowledge and understanding: they are not deducted from a maximum on the basis of errors or omissions.
- (b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.
- (c) Where a wrong answer to part of a question is carried forward and the wrong answer is then used correctly in the following part, give the candidate credit for the subsequent part or 'follow-on'. (GMP 17)
- (d) Award full marks for a correct final answer (including units if required) on its own, unless a numerical question specifically requires evidence of working to be shown, eg in a 'show' question. (GMP 1)
- (e) Award marks where a diagram or sketch conveys correctly the response required by the question. Clear and correct labels (or the use of standard symbols) are usually required for marks to be awarded. (GMP 19)
- (f) Award marks for knowledge of relevant relationships alone. When a candidate writes down several relationships and does not select the correct one to continue with, for example by substituting values, do not award a mark.
- (g) Award marks for non-standard symbols where the symbols are defined and the relationship is correct, or where the substitution shows that the relationship used is correct. This must be clear and unambiguous. (GMP 22)



- (h) Do not award marks if a 'magic triangle' (eg ) is the only statement in a candidate's response. To gain the mark, the correct relationship must be stated, for example  $V = IR$  or  $R = \frac{V}{I}$  (GMP 6)
- (i) In rounding to an expected number of significant figures, award the mark for correct answers which have up to two figures more or one figure less than the number in the data with the fewest significant figures. (GMP 10)  
(Note: the use of a recurrence dot, eg  $0.\dot{6}$ , would imply an infinite number of significant figures and would therefore not be acceptable.)
- (j) Award marks where candidates have incorrectly spelled technical terms, provided that responses can be interpreted and understood without any doubt as to the meaning. Where there is ambiguity, do not award the mark. Two specific examples of this would be when the candidate uses a term that might be interpreted as 'reflection', 'refraction' or 'diffraction' (for example 'defraction'), or one that might be interpreted as either 'fission' or 'fusion' (for example 'fussion'). (GMP 25)

- (k) Only award marks for a valid response to the question asked. Where candidates are asked to:
- **identify, name, give, or state**, they must only name or present in brief form.
  - **describe**, they must provide a statement or structure of characteristics and/or features.
  - **explain**, they must relate cause and effect and/or make relationships between things clear.
  - **determine or calculate**, they must determine a number from given facts, figures or information.
  - **estimate**, they must determine an approximate value for something.
  - **justify**, they must give reasons to support their suggestions or conclusions. For example this might be by identifying an appropriate relationship and the effect of changing variables.
  - **show that**, they must use physics [and mathematics] to prove something, for example a given value - *all steps, including the stated answer, must be shown*.
  - **predict**, they must suggest what may happen based on available information.
  - **suggest**, they must apply their knowledge and understanding of physics to a new situation. A number of responses are acceptable: award marks for any suggestions that are supported by knowledge and understanding of physics.
  - **use their knowledge of physics or aspect of physics to comment on**, they must apply their skills, knowledge and understanding to respond appropriately to the problem/situation presented (for example by making a statement of principle(s) involved and/or a relationship or equation, and applying these to respond to the problem/situation). Candidates gain marks for the breadth and/or depth of their conceptual understanding.

### Standard three marker

The examples over the page set out how to apportion marks to answers requiring calculations. These are the 'standard three marker' type of questions.

Award full marks for a correct answer to a numerical question, even if the steps are not shown explicitly, **unless** it specifically requires evidence of working to be shown.

For some questions requiring numerical calculations, there may be alternative methods (eg alternative relationships) which would lead to a correct answer.

Sometimes, a question requires a calculation which does not fit into the 'standard three marker' type of response. In these cases, the detailed marking instructions will contain guidance for marking the question.

When marking partially correct answers, apportion individual marks as shown over the page.

**Example of a 'standard three marker' question**

The current in a resistor is 1.5 amperes when the potential difference across it is 7.5 volts. Calculate the resistance of the resistor. (3 marks)

| Candidate answer                                                 | Mark and comment                                                       |
|------------------------------------------------------------------|------------------------------------------------------------------------|
| 1. $V = IR$<br>$7.5 = 1.5 \times R$<br>$R = 5.0 \Omega$          | 1 mark: relationship<br>1 mark: substitution<br>1 mark: correct answer |
| 2. $5.0 \Omega$                                                  | 3 marks: correct answer                                                |
| 3. $5.0$                                                         | 2 marks: unit missing                                                  |
| 4. $4.0 \Omega$                                                  | 0 marks: no evidence, wrong answer                                     |
| 5. $\_\_\Omega$                                                  | 0 marks: no working or final answer                                    |
| 6. $R = \frac{V}{I} = \frac{7.5}{1.5} = 4.0 \Omega$              | 2 marks: arithmetic error                                              |
| 7. $R = \frac{V}{I} = 4.0 \Omega$                                | 1 mark: relationship only                                              |
| 8. $R = \frac{V}{I} = \_\_\Omega$                                | 1 mark: relationship only                                              |
| 9. $R = \frac{V}{I} = \frac{7.5}{1.5} = \_\_\Omega$              | 2 marks: relationship and substitution, no final answer                |
| 10. $R = \frac{V}{I} = \frac{7.5}{1.5} = 4.0$                    | 2 marks: relationship and substitution, wrong answer                   |
| 11. $R = \frac{V}{I} = \frac{1.5}{7.5} = 5.0 \Omega$             | 1 mark: relationship but wrong substitution                            |
| 12. $R = \frac{V}{I} = \frac{75}{1.5} = 5.0 \Omega$              | 1 mark: relationship but wrong substitution                            |
| 13. $R = \frac{I}{V} = \frac{7.5}{1.5} = 5.0 \Omega$             | 0 marks: wrong relationship                                            |
| 14. $V = IR$<br>$7.5 = 1.5 \times R$<br>$R = 0.2 \Omega$         | 2 marks: relationship and substitution, arithmetic error               |
| 15. $V = IR$<br>$R = \frac{I}{V} = \frac{1.5}{7.5} = 0.2 \Omega$ | 1 mark: relationship only, wrong rearrangement of symbols              |

# Marking instructions for each question

| Question |     |      | Expected response                                                                                      | Max mark | Additional guidance                                                                                                                                                                                                                   |
|----------|-----|------|--------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.       | (a) | (i)  | $(v_h = 16 \cdot 0 \cos 42 \cdot 0)$<br>$v_h = 11 \cdot 9 \text{ m s}^{-1}$                            | 1        | Accept: 12, 11.89, 11.890                                                                                                                                                                                                             |
|          |     | (ii) | $(v_v = 16 \cdot 0 \sin 42 \cdot 0)$<br>$v_v = 10 \cdot 7 \text{ m s}^{-1}$                            | 1        | Accept: 11, 10.71, 10.706                                                                                                                                                                                                             |
|          | (b) |      | $v = u + at$ (1)<br>$0 = 10 \cdot 7 + (-9 \cdot 8)t$ (1)<br>$t = 1 \cdot 1 \text{ s}$ (1)              | 3        | Or consistent with (a)(ii)<br>$u$ and $a$ must have opposite signs<br>Accept: 1, 1.09, 1.092<br><br>For alternative methods:<br>1 mark for <b>all</b> relationships<br>1 mark for <b>all</b> substitutions<br>1 mark for final answer |
|          | (c) |      | $s = vt$ (1)<br>$s = 11 \cdot 9 \times (1 \cdot 1 + 1 \cdot 40)$ (1)<br>$s = 29 \cdot 8 \text{ m}$ (1) | 3        | Or consistent with (a)(i) and (b)<br>Accept: 29.75, 29.750<br>Also accept 30                                                                                                                                                          |
|          | (d) |      | Greater (1)<br><br>The skier has a greater speed/<br>velocity as they land. (1)                        | 2        | Potential energy at take-off is<br>transferred/converted to kinetic<br>energy.                                                                                                                                                        |



| Question |     |      | Expected response                                                                                                                                                                                                                                | Max mark | Additional guidance                                                                                                                                                                                                 |
|----------|-----|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.       | (a) |      | $F = ma$ (1)<br>$1.15 \times 10^5 = (9.75 \times 10^4 + 3.56 \times 10^4) \times a$ (1)<br><br>$(F = ma)$<br>$F = 3.56 \times 10^4 \times \left( \frac{1.15 \times 10^5}{1.331 \times 10^5} \right)$ (1)<br>$F = 3.08 \times 10^4 \text{ N}$ (1) | 4        | Accept 3.1, 3.076, 3.0759<br><br>$F = ma$ anywhere, 1 mark                                                                                                                                                          |
|          | (b) | (i)  | $f_0 = f_s \left( \frac{v}{v \pm v_s} \right)$ (1)<br>$531 = 511 \left( \frac{340}{340 - v_s} \right)$ (1)<br>$v_s = 13 \text{ m s}^{-1}$ (1)                                                                                                    | 3        | Accept $f_0 = f_s \left( \frac{v}{v - v_s} \right)$<br>Accept 10, 12.8, 12.81                                                                                                                                       |
|          |     | (ii) | Not correct/incorrect (1)<br><br>The passenger and engine are travelling at the same velocity. (1)                                                                                                                                               | 2        | MUST JUSTIFY<br><br>Accept:<br>The passenger is travelling at the <u>same speed and in the same direction</u> as the whistle/engine.<br><br>The distance between the whistle/engine and passenger remains constant. |

| Question |     |  | Expected response                                                                                                                            | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-----|--|----------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 3.       | (a) |  | $v^2 = u^2 + 2as$ (1)<br>$v^2 = 0^2 + 2 \times (-)9.8 \times (-)1.27$ (1)<br>$v = 5.0 \text{ m s}^{-1}$ (1)                                  | 3        | Accept: 5, 4.99, 4.989<br><i>a</i> and <i>s</i> <b>must</b> have the same sign, otherwise max 1 mark.<br><br>For alternative methods:<br>1 mark for <b>all</b> relationships<br>1 mark for <b>all</b> substitutions<br>1 mark for final answer<br>eg<br>$E_p = mgh$<br>$E_p = 1.59 \times 10^{-2} \times 9.8 \times 1.27$<br>$E_k = \frac{1}{2}mv^2$<br>$(1.59 \times 10^{-2} \times 9.8 \times 1.27) = \frac{1}{2} \times 1.59 \times 10^{-2} \times v^2$<br>$v = 5.0 \text{ ms}^{-1}$ |
|          | (b) |  | $Ft = mv - mu$ (1)<br>$0.14 = (1.59 \times 10^{-2} \times v) - (1.59 \times 10^{-2} \times -5.0)$ (1)<br>$v = 3.8 \text{ m s}^{-1}$ (1)      | 3        | Or consistent with (a)<br>Accept: 4, 3.81, 3.805<br><i>Ft</i> and <i>u</i> must have opposite signs otherwise max 1 mark.<br><br>Accept:<br>$\Delta p = mv - mu$<br>$p = mv$<br><br>Do not accept $p = mv - mu$                                                                                                                                                                                                                                                                         |
|          | (c) |  | <u>Kinetic energy</u> is greater before (the collision) than after.<br><br><b>OR</b><br><u>Kinetic energy</u> is lost (during the collision) | 1        | Do not accept $E_k$ before not equal to $E_k$ after.<br><br>Do not accept $E_k$ is not conserved.                                                                                                                                                                                                                                                                                                                                                                                       |
|          | (d) |  | (Softer material would) increase the time of contact (1)<br>and decrease the (maximum/average) force (1)                                     | 2        | Independent marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

| Question |  |  | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Max mark | Additional guidance                                                                                                                                                                            |
|----------|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 4.       |  |  | <p><b>Award 3 marks</b> where the candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.</p> <p><b>Award 2 marks</b> where the candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.</p> <p><b>Award 1 mark</b> where the candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.</p> <p><b>Award 0 marks</b> where the candidate has not demonstrated an understanding of the physics involved. There is no evidence that they have recognised the area of physics involved, or they have not given any statement of a relevant physics principle. Award this mark also if the candidate merely restates the physics given in the question.</p> | 3        | <p>Candidates may use a variety of physics arguments to answer this question.</p> <p>Award marks based on candidates demonstrating overall good, reasonable, limited, or no understanding.</p> |

| Question |     |      | Expected response                                                                                                                                                                                               | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|----------|-----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.       | (a) |      | $F = G \frac{m_1 m_2}{r^2} \quad (1)$ $1.59 \times 10^{39} = 6.67 \times 10^{-11} \times \frac{3.18 \times 10^{30} \times 2.27 \times 10^{30}}{r^2} \quad (1)$ $r = 5.50 \times 10^5 \text{ m} \quad (1)$       | 3        | Accept: 5.5, 5.503, 5.5029                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|          | (b) | (i)  | <p>Waves <u>meet</u><br/> 180°/completely/totally/exactly out of phase<br/> <b>OR</b><br/> Crest <u>meets</u> trough<br/> <b>OR</b><br/> Path difference = <math>\left(m + \frac{1}{2}\right)\lambda</math></p> | 1        | Can be shown by appropriate diagram                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|          |     | (ii) | $\left(\frac{4.0 \times 10^{-18}}{4.0 \times 10^3}\right) 10^{-21} \quad (1)$ <p>(change in length is) <u>21</u> orders of magnitude <u>smaller</u> <span style="float:right">(1)</span></p>                    | 2        | <p>Accept <math>\left(\frac{10^{-18}}{10^3}\right) 10^{-21}</math></p> <p><b>OR</b></p> <p><math>(-18-3) = -21 \quad (1)</math></p> <p>Accept<br/> 21 <u>smaller</u> on its own <span style="float:right">(2)</span></p> <p>Do not accept<br/> 21 <u>times</u> smaller on its own (0)</p> <p>Accept <math>\left(\frac{10^3}{10^{-18}}\right) 10^{21}</math></p> <p><b>OR</b></p> <p><math>3-(-18) = 21 \quad (1)</math></p> <p>Accept: the length of the arm is 21 orders of magnitude greater than the change in length. <span style="float:right">(1)</span></p> |

| Question |     |       | Expected response                                                                                                                                                 | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                                                                        |
|----------|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 6.       | (a) | (i)   | $E_2 - E_1 = hf$ (1)<br>$(-0.871 \times 10^{-19} - (-5.45 \times 10^{-19})) =$ (1)<br>$6.63 \times 10^{-34} \times f$<br>$f = 6.91 \times 10^{14} \text{ Hz}$ (1) | 3        | Accept: 6.9, 6.906, 6.9065<br><br>Accept:<br>$E_1 - E_4 = -hf$<br>$E_4 - E_1 = hf$<br>$(\Delta) E = hf$<br>for relationship mark anywhere<br><br>Accept:<br>$(5.45 \times 10^{-19} - 0.871 \times 10^{-19}) =$<br>$6.63 \times 10^{-34} \times f$<br><br>If $(0.871 \times 10^{-19} - 5.45 \times 10^{-19})$ shown<br>for substitution, maximum 1 mark<br>for relationship |
|          |     | (ii)  | $v = f\lambda$ (1)<br>$3.00 \times 10^8 = 6.91 \times 10^{14} \times \lambda$ (1)<br>$\lambda = 4.34 \times 10^{-7} \text{ m}$ (1)                                | 3        | Or consistent with (a)(i)<br><br>Accept: 4.3, 4.342, 4.3415                                                                                                                                                                                                                                                                                                                |
|          |     | (iii) | Blue-violet                                                                                                                                                       | 1        | Or consistent with (a)(ii)                                                                                                                                                                                                                                                                                                                                                 |
|          | (b) |       | $z = \frac{v}{c}$ (1)<br>$z = \frac{4.51 \times 10^6}{3.00 \times 10^8}$ (1)<br>$z = 0.0150$ (1)                                                                  | 3        | Accept: 0.015, 0.01503, 0.015033                                                                                                                                                                                                                                                                                                                                           |
|          | (c) |       | Redshift is evidence that the<br>Universe is expanding (1)<br><br><u>Expanding Universe</u> is evidence<br>supporting the Big Bang theory (1)                     | 2        | Accept: Redshift is evidence that the<br>galaxies are moving away from each<br>other.                                                                                                                                                                                                                                                                                      |

| Question |     |      | Expected response                                                                                                                                                                                                                                                                                         | Max mark | Additional guidance                                                                                                                                                                      |
|----------|-----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 7.       | (a) | (i)  | To ensure the (accelerating) force on the hydrogen ion is in the same direction.<br><br><b>OR</b><br><br>To ensure the hydrogen ions accelerate in the same direction.<br><br><b>OR</b><br><br>To ensure that the direction of the electric field is correct when the hydrogen ions pass across the gaps. | 1        | Response must make some implication of 'same direction'.                                                                                                                                 |
|          |     | (ii) | As the speed of hydrogen ions increases, they travel further in the same time.                                                                                                                                                                                                                            | 1        | Accept:<br>So that the hydrogen ions are at the ends of the tubes when the field changes polarity.<br><b>OR</b><br>So that a constant frequency AC supply can be used.                   |
|          | (b) |      | $l' = l \sqrt{1 - \left(\frac{v}{c}\right)^2} \quad (1)$ $l' = 13 \sqrt{1 - \left(\frac{0.50c}{c}\right)^2} \quad (1)$ $l' = 11 \text{ m} \quad (1)$                                                                                                                                                      | 3        | Accept: 10, 11.3, 11.26<br><br>Alternative substitutions:<br><br>$l' = 13 \sqrt{1 - (0.50)^2}$ $l' = 13 \sqrt{1 - \left(\frac{0.50 \times 3.00 \times 10^8}{3.00 \times 10^8}\right)^2}$ |
|          | (c) | (i)  | A (composite) particle made of a quark-antiquark pair.                                                                                                                                                                                                                                                    | 1        | Do not accept: made of two quarks                                                                                                                                                        |
|          |     | (ii) | Into the page                                                                                                                                                                                                                                                                                             | 1        |                                                                                                                                                                                          |
|          | (d) | (i)  | W boson<br><br><b>OR</b><br><br>Z boson                                                                                                                                                                                                                                                                   | 1        |                                                                                                                                                                                          |
|          |     | (ii) | $4.20 \text{ GeV} = 4.20 \times 10^9 \times 1.60 \times 10^{-19} \quad (1)$ $E = mc^2 \quad (1)$ $(4.20 \times 10^9 \times 1.60 \times 10^{-19}) = m \times (3.00 \times 10^8)^2 \quad (1)$ $m = 7.47 \times 10^{-27} \text{ kg} \quad (1)$                                                               | 4        | Accept: 7.5, 7.467, 7.4667<br><br>Relationship anywhere 1 mark.                                                                                                                          |

| Question |     |       | Expected response                                                                                                                                                                                                                                                                                                        | Max mark | Additional guidance                                                                                    |
|----------|-----|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------|
| 8.       | (a) |       | <p>The frequency of the UV is greater than the threshold frequency, whereas the frequency of white light is less than the threshold frequency.</p> <p><b>OR</b></p> <p>The energy of a photon of UV is greater than the work function, whereas the energy of a photon of white light is less than the work function.</p> | 1        | Response must refer to both UV and white light.                                                        |
|          | (b) | (i)   | $1.1 \times 10^{-19} \text{ J}$                                                                                                                                                                                                                                                                                          | 1        |                                                                                                        |
|          |     | (ii)  | $W = QV$ (1)<br>$W = 1.60 \times 10^{-19} \times 12.0$ (1)<br>$W = 1.92 \times 10^{-18} \text{ J}$                                                                                                                                                                                                                       | 2        | SHOW                                                                                                   |
|          |     | (iii) | $E_k = 1.1 \times 10^{-19} + 1.92 \times 10^{-18}$ (1)<br>$E_k = \frac{1}{2}mv^2$ (1)<br>$(1.1 \times 10^{-19} + 1.92 \times 10^{-18}) =$ (1)<br>$\frac{1}{2} \times 9.11 \times 10^{-31} \times v^2$<br>$v = 2.11 \times 10^6 \text{ m s}^{-1}$ (1)                                                                     | 4        | <p>Or consistent with (b)(i)</p> <p>Accept: 2.1, 2.111, 2.1111</p> <p>Relationship anywhere 1 mark</p> |

| Question |     |              | Expected response                                                                                                                                                                                                                                                                          | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                                                                             |
|----------|-----|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9.       | (a) | (i)          | $F = 19.5 \sin 14.0$ (1)<br>$F_R = (2 \times 19.5 \sin 14.0) = 9.43 \text{ N}$ (1)<br><b>OR</b><br>$F_R = 2 \times 19.5 \sin 14.0$ (1)<br>$F_R = 9.43 \text{ N}$ (1)                                                                                                                       | 2        | Accept: 9.4, 9.435, 9.4350<br><br>Or by scale diagram:<br>1 for suitable scale diagram<br>1 for correct answer                                                                                                                                                                                                                                                                  |
|          |     | (ii)         | No resultant force in this direction/<br>the sideways direction<br><b>OR</b><br>Unbalanced force in this direction/<br>the sideways direction is 0 N<br><b>OR</b><br>The components of the force at $90^\circ$<br>to the direction of the movement<br>are equal and opposite/balanced. (1) | 1        | Accept reference to horizontal<br>forces/left and right direction, since<br>the diagram orientation makes it<br>clear which forces are being referred<br>to.<br><br>Do not accept: 'the forces are<br>balanced' alone                                                                                                                                                           |
|          | (b) | (i)          | $I = \frac{P}{A}$ (1)<br>$11800 = \frac{P}{1.24 \times 10^{-5}}$ (1)<br>$P = \frac{E}{t}$ (1)<br>$(11800 \times 1.24 \times 10^{-5}) = \frac{2.10}{t}$ (1)<br>$t = 14.4 \text{ s}$ (1)                                                                                                     | 5        | Accept: 14, 14.35, 14.352<br><br>$I = \frac{P}{A}$ anywhere, 1 mark<br>$P = \frac{E}{t}$ anywhere, 1 mark                                                                                                                                                                                                                                                                       |
|          |     | (ii)         | $6.3 \times 0.30^2 = 0.57$<br>$3.5 \times 0.40^2 = 0.56$<br>$2.3 \times 0.50^2 = 0.58$<br>$1.6 \times 0.60^2 = 0.58$ (2)<br><br>Statement of $I \times d^2 = \text{constant}$ , so<br>LED is a point source (1)                                                                            | 3        | All four calculations correct (2)<br>Three calculations correct (1)<br><Three calculations correct (0)<br><br>This conclusion mark is only<br>available if consistent with the<br>calculations shown.<br><br>Graphical method:<br><br>Graph drawn correctly (1)<br>Best fit line through origin (1)<br>Statement of $I \propto \frac{1}{d^2}$ , so LED is a<br>point source (1) |
|          |     | (iii)<br>(A) | A semiconductor that has (specific)<br>impurities added                                                                                                                                                                                                                                    | 1        |                                                                                                                                                                                                                                                                                                                                                                                 |



| Question |  |     | Expected response                                                                                                                                                                                                                                                                                                   | Max mark | Additional guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|--|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|          |  | (B) | <p>(Voltage applied causes) electrons to move from the conduction band of the n-type (semiconductor) towards the conduction band of the p-type (semiconductor). (1)</p> <p>Electrons ‘fall’ from the conduction band into the valence band (on either side of the junction) (1)</p> <p>Photons are emitted. (1)</p> | 3        | <p>Any answer using recombination of holes and electrons <b>on its own</b>, with no reference to band theory, is worth 0 marks</p> <p>Any wrong physics eg holes move up (from valence band to conduction band)- 0 marks</p> <p>To access this mark, the direction the electrons move must be clear.</p> <p>To access this mark, valence and conduction bands must be included in the answer.</p> <p>Do not accept: ‘valency’ as a name for the valence band or ‘conductive’ as a name for the conduction band.</p> <p>This mark is dependent upon having at least one of the first two statements.</p> |

| Question |     |       | Expected response                                                                                                                         | Max mark | Additional guidance                                                                                                                                                            |
|----------|-----|-------|-------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10.      | (a) | (i)   | (A) $13.9^\circ$                                                                                                                          | 1        | Do not accept:14                                                                                                                                                               |
|          |     |       | (B) $\left( \Delta R = \frac{R_{\max} - R_{\min}}{n} \right)$<br>$\Delta R = \frac{14.5 - 13.0}{5}$ (1)<br>$\Delta R = 0.3^\circ$ (1)     | 2        |                                                                                                                                                                                |
|          |     | (ii)  | $m\lambda = d \sin \theta$ (1)<br>$2 \times \lambda = 4.00 \times 10^{-6} \sin 13.9$ (1)<br>$\lambda = 4.80 \times 10^{-7} \text{ m}$ (1) | 3        | Or consistent with (a)(i)(A)<br>Accept: 4.8, 4.805, 4.8046                                                                                                                     |
|          |     | (iii) | Percentage (scale reading) uncertainty in the angle is smaller(1)                                                                         | 1        | Accept: fractional uncertainty in place of percentage uncertainty<br><br>Must be percentage or fractional uncertainty not just scale reading uncertainty or uncertainty alone. |
|          | (b) |       | The path difference (at the central maximum) for each wavelength/ frequency/ colour will be zero (1)                                      | 1        | Must answer in terms of path difference.                                                                                                                                       |

| Question |  |  | Expected response                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Max mark | Additional guidance                                                                                                                                                                            |
|----------|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11.      |  |  | <p><b>Award 3 marks</b> where the candidate has demonstrated a good understanding of the physics involved. They show a good comprehension of the physics of the situation and provide a logically correct answer to the question posed. This type of response might include a statement of the principles involved, a relationship or an equation, and the application of these to respond to the problem. The answer does not need to be 'excellent' or 'complete' for the candidate to gain full marks.</p> <p><b>Award 2 marks</b> where the candidate has demonstrated a reasonable understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood the problem.</p> <p><b>Award 1 mark</b> where the candidate has demonstrated a limited understanding of the physics involved. They make some statement(s) that are relevant to the situation, showing that they have understood at least a little of the physics within the problem.</p> <p><b>Award 0 marks</b> where the candidate has not demonstrated an understanding of the physics involved. There is no evidence that they have recognised the area of physics involved, or they have not given any statement of a relevant physics principle. Award this mark also if the candidate merely restates the physics given in the question.</p> | 3        | <p>Candidates may use a variety of physics arguments to answer this question.</p> <p>Award marks based on candidates demonstrating overall good, reasonable, limited, or no understanding.</p> |

| Question |     |      | Expected response                                                                                                                                         | Max mark | Additional guidance                                                                            |
|----------|-----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------|
| 12.      | (a) | (i)  | $n = \frac{\sin \theta_1}{\sin \theta_2} \quad (1)$ $n = \frac{\sin 47.0}{\sin 31.0} \quad (1)$ $n = 1.42 \quad (1)$                                      | 3        | Also accept 1.4, 1.420, 1.4200                                                                 |
|          |     | (ii) | (frequency is the) same                                                                                                                                   | 1        |                                                                                                |
|          | (b) |      | Ray drawn at smaller angle of refraction                                                                                                                  | 1        | Ignore any emergent rays<br>Ray must be passably straight.                                     |
|          | (c) |      | <p>green light has a higher/larger/greater frequency (1)</p> <p>so the refractive index is greater (and the ray refracts more/at a smaller angle) (1)</p> | 2        | Any mention of a greater angle of refraction or no change in the angle of refraction - 0 marks |

| Question |     |      | Expected response                                                                                                                                                                             | Max mark | Additional guidance                              |
|----------|-----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|--------------------------------------------------|
| 13.      | (a) |      | (An alternating current) <u>changes direction and</u> (instantaneous) <u>value with time.</u>                                                                                                 | 1        |                                                  |
|          | (b) | (i)  | $(V_{peak} = 5.0 \times 3)$<br><br>$V_{peak} = 15 \text{ V}$                                                                                                                                  | 1        |                                                  |
|          |     | (ii) | $(T = 1.0 \times 10^{-3} \times 4 = 4.0 \times 10^{-3} \text{ s})$<br><br>$f = \frac{1}{T} \text{ (1)}$<br><br>$f = \frac{1}{4.0 \times 10^{-3}} \text{ (1)}$<br><br>$f = 250 \text{ Hz (1)}$ | 3        |                                                  |
|          | (c) |      | Same frequency and peak voltage (1)<br><br>Trace shows 'half-wave rectification' (1)                                                                                                          | 2        | Positive or negative half of the cycle accepted. |

| Question |     |      | Expected response                                                                                                                              | Max mark | Additional guidance                                                                                                                                                                          |
|----------|-----|------|------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 14.      | (a) |      | Adjust variable resistor and take readings of $V$ and $I$ . (1)<br>Plot a graph of $V$ against $I$ . (1)<br>Gradient of graph = $-r$ . (1)     | 3        | Measure open circuit voltage $E$ /measure the voltage $E$ when the switch is open. (1)<br>Close the switch and take a reading of $V$ and $I$ . (1)<br>Calculate $r$ using $E = V + Ir$ . (1) |
|          | (b) | (i)  | 1.5 J of energy is supplied to/gained by each coulomb (of charge passing through the cell).                                                    | 1        |                                                                                                                                                                                              |
|          |     | (ii) | $E = V + Ir$ and $V = IR$ (1)<br>$6.0 = (0.20R + (0.20 \times 2.0))$ (1)<br>$R = 28 \Omega$ (1)<br>$(R_v = 28 - 20)$<br>$R_v = 8.0 \Omega$ (1) | 4        | Accept: $E = I(R + r)$<br>Accept: 8, 8.00, 8.000                                                                                                                                             |
|          | (c) |      | Increases (1)<br>Current is less (1)<br>Lost volts ( $Ir$ ) decreases (1)                                                                      | 3        | Look for this statement first - if incorrect or missing then (0 marks).                                                                                                                      |

| Question |     |       | Expected response                                                                                                                                                                  | Max mark | Additional guidance                                                                                                                                                                                                                                                                           |
|----------|-----|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 15.      | (a) |       | <p>The frictional force/drag acting on the ball bearing increases (as its speed increases). (1)</p> <p>The <u>frictional force/drag</u> and <u>weight</u> become balanced. (1)</p> | 2        |                                                                                                                                                                                                                                                                                               |
|          | (b) | (i)   | <p>Appropriate labels and units (1)</p> <p>Suitable scales (1)</p> <p>Correct plotting of points and appropriate line of best fit (1)</p>                                          | 3        | <p>Allow for axes starting at zero or broken axes or at an appropriate value.</p> <p>Accuracy of plotting should be easily checkable with the scale chosen.</p> <p>Do not penalise if the candidate plots <math>d^2</math> against <math>v_t</math>.</p>                                      |
|          |     | (ii)  | <p>There is a non-zero y-intercept/<br/>The line of best fit does not go through the origin</p>                                                                                    | 1        |                                                                                                                                                                                                                                                                                               |
|          |     | (iii) | <p><math>m = \frac{y_2 - y_1}{x_2 - x_1}</math> (1)</p> <p>Correctly calculated gradient (1)</p>                                                                                   | 2        | <p><b>Must</b> be consistent with graph drawn for (i).</p> <p>Candidates are asked to calculate the gradient of <b>their graph</b>.</p> <p>Unit not required but if a unit is given it must be correct.</p> <p>Tolerance required depending upon line of best fit drawn by the candidate.</p> |
|          |     | (iv)  | <p><math>m = \frac{375g}{\eta}</math> (1)</p> <p>Correctly calculated viscosity consistent with b(iii), including correct unit. (1)</p>                                            | 2        |                                                                                                                                                                                                                                                                                               |

[END OF MARKING INSTRUCTIONS]

The following table provides information on each question including: Course content being assessed, Skills assessed (see Physics Understanding Standards materials for a definition of each code); Maximum Mark; A-type marks.

| Question | Part     | Course content                                             | Skills assessed | Maximum mark | A-type marks |
|----------|----------|------------------------------------------------------------|-----------------|--------------|--------------|
| 1        | (a)(i)   | Our dynamic Universe - gravitation                         | K3              | 1            |              |
|          | (a)(ii)  | Our dynamic Universe - gravitation                         | K3              | 1            |              |
|          | (b)      | Our dynamic Universe - motion-equations and graphs         | K3              | 3            |              |
|          | (c)      | Our dynamic Universe - motion-equations and graphs         | K3              | 3            | 2            |
|          | (d)      | Our dynamic Universe - forces, energy, and power           | K2              | 2            |              |
| 2        | (a)      | Our dynamic Universe - forces, energy, and power           | K3              | 4            |              |
|          | (b)(i)   | Our dynamic Universe - expanding Universe                  | K3              | 3            |              |
|          | (b)(ii)  | Our dynamic Universe - expanding Universe                  | S6              | 2            | 2            |
| 3        | (a)      | Our dynamic Universe - motion-equations and graphs         | K3              | 3            |              |
|          | (b)      | Our dynamic Universe - collisions, explosions, and impulse | K3              | 3            |              |
|          | (c)      | Our dynamic Universe - collisions, explosions, and impulse | K2              | 1            |              |
|          | (d)      | Our dynamic Universe - collisions, explosions, and impulse | K2              | 2            |              |
| 4        |          | Our dynamic Universe - expanding Universe                  | K2              | 3            | 2            |
| 5        | (a)      | Our dynamic Universe - gravitation                         | K3              | 3            |              |
|          | (b)(i)   | Particles and waves - interference                         | K2              | 1            |              |
|          | (b)(ii)  | Particles and waves - Standard Model                       | S4              | 2            | 1            |
| 6        | (a)(i)   | Particles and waves - spectra                              | K3              | 3            |              |
|          | (a)(ii)  | Particles and waves - spectra                              | K3              | 3            |              |
|          | (a)(iii) | Particles and waves - spectra                              | S2              | 1            |              |
|          | (b)      | Our dynamic Universe - expanding Universe                  | K3              | 3            |              |
|          | (c)      | Our dynamic Universe - expanding Universe                  | K2              | 2            | 1            |
| 7        | (a)(i)   | Particles and waves - forces on charged particles          | K2              | 1            | 1            |
|          | (a)(ii)  | Particles and waves - forces on charged particles          | S6              | 1            | 1            |
|          | (b)      | Our dynamic Universe - special relativity                  | K3              | 3            |              |
|          | (c)(i)   | Particles and waves - Standard Model                       | K1              | 1            |              |
|          | (c)(ii)  | Particles and waves - forces on charged particles          | S6              | 1            |              |
|          | (d)(i)   | Particles and waves - Standard Model                       | K1              | 1            |              |
|          | (d)(ii)  | Particles and waves - nuclear reactions                    | S4              | 1            | 1            |
|          |          |                                                            | K3              | 3            | 2            |
| 8        | (a)      | Particles and waves - wave-particle duality                | K2              | 1            | 1            |
|          | (b)(i)   | Particles and waves - wave-particle duality                | S4              | 1            |              |
|          | (b)(ii)  | Particles and waves - forces on charged particles          | K3              | 2            |              |
|          | (b)(iii) | Particles and waves - forces on charged particles          | K3              | 4            | 3            |



|    |             |                                                          |    |   |   |
|----|-------------|----------------------------------------------------------|----|---|---|
| 9  | (a)(i)      | Our dynamic Universe - forces, energy, and power         | K3 | 2 | 1 |
|    | (a)(ii)     | Our dynamic Universe - forces, energy, and power         | K2 | 1 | 1 |
|    | (b)(i)      | Particles and waves - inverse square law                 | K3 | 5 | 3 |
|    | (b)(ii)     | Particles and waves - inverse square law                 | S4 | 2 |   |
|    |             |                                                          | S6 | 1 | 1 |
|    | (b)(iii)(A) | Electricity - semiconductors and p-n junctions           | K1 | 1 |   |
|    | (b)(iii)(B) | Electricity - semiconductors and p-n junctions           | K2 | 3 | 3 |
| 10 | (a)(i)(A)   | Uncertainties                                            | S4 | 1 |   |
|    | (a)(i)(B)   | Uncertainties                                            | K3 | 2 |   |
|    | (a)(ii)     | Particles and waves - interference                       | K3 | 3 |   |
|    | (a)(iii)    | Uncertainties                                            | S7 | 1 | 1 |
|    | (b)         | Particles and waves - interference                       | K2 | 1 | 1 |
| 11 |             | Particles and waves - spectra                            | K2 | 3 | 2 |
| 12 | (a)(i)      | Particles and waves - refraction of light                | K3 | 3 |   |
|    | (a)(ii)     | Particles and waves - refraction of light                | K1 | 1 |   |
|    | (b)         | Particles and waves - refraction of light                | S3 | 1 |   |
|    | (c)         | Particles and waves - refraction of light                | K2 | 2 | 2 |
| 13 | (a)         | Electricity - monitoring and measuring AC                | K1 | 1 |   |
|    | (b)(i)      | Electricity - monitoring and measuring AC                | S4 | 1 |   |
|    | (b)(ii)     | Electricity - monitoring and measuring AC                | S4 | 3 |   |
|    | (c)         | Electricity - monitoring and measuring AC                | S5 | 2 | 1 |
| 14 | (a)         | Electricity - electrical sources and internal resistance | S1 | 3 | 1 |
|    | (b)(i)      | Electricity - electrical sources and internal resistance | K1 | 1 |   |
|    | (b)(ii)     | Electricity - electrical sources and internal resistance | K3 | 4 | 1 |
|    | (c)         | Electricity - electrical sources and internal resistance | K2 | 3 | 2 |
| 15 | (a)         | Our dynamic Universe - forces, energy, and power         | K2 | 2 | 1 |
|    | (b)(i)      | Unfamiliar content - skills                              | S3 | 3 |   |
|    | (b)(ii)     | Unfamiliar content - skills                              | S6 | 1 |   |
|    | (b)(iii)    | Unfamiliar content - skills                              | S4 | 2 |   |
|    | (b)(iv)     | Unfamiliar content - skills                              | S4 | 2 | 2 |